



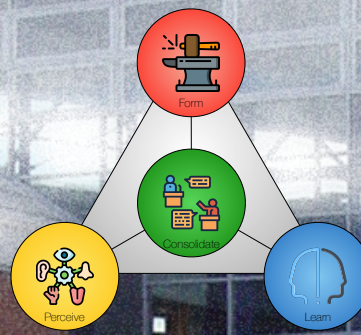
AALBORG UNIVERSITY
DENMARK

COURSE OVERVIEW

RESUME OF 12 LECTURES
MINIPROJECTS
EXAM

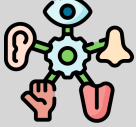
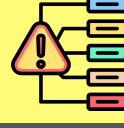
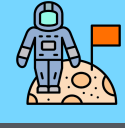
IVAN AAEN
INSTITUT FOR DATALOGI
IVAN@CS.AAU.DK

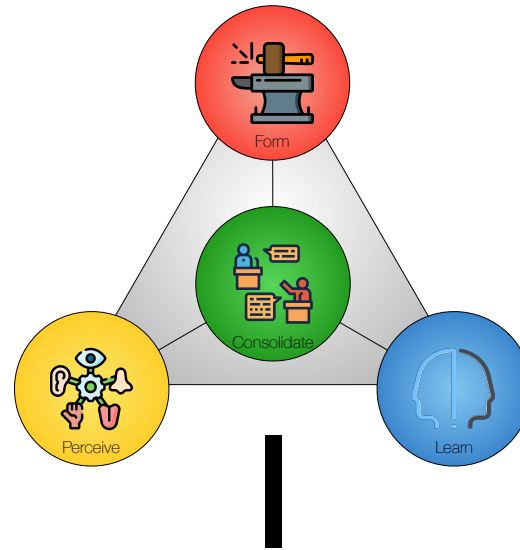
SWI 2026



VIT: Problem-Solution Canvas

Canvas as a **configuration** — a **status** over ideas and principles

	 PERCEIVE	 FORM	 CONSOLIDATE	 LEARN
 RATIONALE	 Problem Scenarios	 Leverage Scenarios	 Solution Scenarios	 Horizon H3
 STRATEGY	 Outer VUCA Strategies	 Inner VUCA Strategies	 Evolvability Diffuse, Adopt	 Horizon H2
 TACTICS	 Manifestations Abstraction	 Capabilities Abstraction	 Value Func/Non-func	 Horizon H1



Roles and Core Activities

Exercise 0 (Innovation opportunities)

Exercise 0 is all about innovation opportunities for you this semester.

Look at the semester project description in the study regulations for your education. See if your topic is related to one or more of the following trends and forces:

- Do you see trends in **technology** that might be related to your topic?
- Do you see trends in **legislation**, **society**, **culture**, and **economy** that might be linked to your topic?
- Do you see new **forces** in industry and markets, new **needs** and **demands**, new **products**, and new **conditions** relevant to your topic?
- How can such trends and forces create **innovation opportunities** in your project?

Exercise 1 (Innovation in your project)

Exercise 1 is a follow-up to Exercise 0. In this exercise, we will consider your semester project:

- What **problem** are you working on?
- Explain what **type** of innovation you aim for (Table 1.1 in Essence).
- What will be the **keystones** in your design? What will make your design stand out?
- Who will **benefit** from it? What **value** will you create?
- Where will it be **used**? How will it create **change**?
- Characterize your problem or solution challenges in terms of **volatility**, **uncertainty**, **complexity**, and **ambiguity** (Section 1.2 in Essence).
- Do you consider your project to be mainly **demand-pull** or **technology-push**? (Section 1.2 in Essence)
- Will **roles** be relevant? How? When? What are the **pros** and **cons** of roles?

Digital Innovation Defined

*Products, services, processes, or business models that emerge **within** or **between** business units and where **digital technology is a key factor** in triggering or enabling value.*

Four Types of Digital Innovation

<i>Main definition</i>		Products, services, processes, or business models that emerge within or between business units and where digital technology is a key factor in triggering or enabling value
<i>Variants</i>	<i>Product</i>	New or radically changed products or services
	<i>Process</i>	New or radically improved ways to produce products or services
	<i>Position</i>	Products, services, processes, or business models that emerge by fitting earlier solutions into new uses
	<i>Paradigm</i>	Products, services, processes, or business models that emerge from radical changes in the mental models of what a business is; who the users are; or what the market is

Table 1.1: Four Types of Digital Innovation

We need a Third Paradigm

1.The document-oriented.

- Top-down, static, plan-oriented.

2.The agile.

- Incremental, dynamic, user-oriented.

3.The pragmatic.

- Problem- and solution-oriented.

Factors of Change

- The Anthropocene.
- Knightian uncertainties (and opportunities).
- Hypercomplexity.
- Problems and solutions are interdependent.
- Design takes place in a VUCA world.

Hypercomplexity

Hypercomplexity is a degree of complexity which inhibits or makes impossible rational decision making within reasonable time constraints (Bohman, 1995).

- Systems become increasingly interdependent with open system boundaries.
- Increasingly, digital ecologies emerge while we design.
- New components, sensors, data, platforms, and mobility change the system ecology dynamically.
- Radical increases in what means are available.
- Systems become increasingly intelligent.

Problems and Solutions

Shape Each Other

- We refer to what we are building as the *Inner Environment*.
 - The Inner Environment denotes the substance and organization of our contributions.
- We refer to what we are building this for as the *Outer Environment*.
 - The Outer Environment denotes the surroundings where our contributions will reside and operate.
- Both environments involve choice.

Roles are Structural Concepts in Essence



Child

Any team member exploring the *situation*. Unorthodox and creative. Offers ideas and proposals without responsibility. Questions conventional views on uses, needs, users, and the application of technology. This Role is fleeting.



Responder

Any developer working on the *contribution*. Offers technological responses to challenges—always with a view to technological options and affordances that come up during the design process.



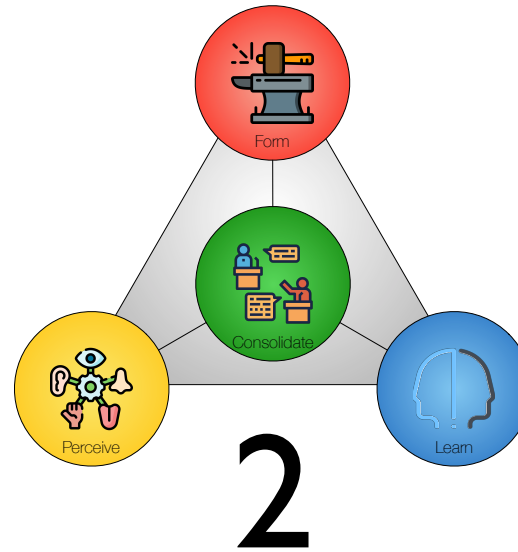
Challenger

The team member who provides the reasoning and politics underlying the *solution*. Aiming for a solution with a good balance between problem and contribution in the specific situation. Open to new ideas and good arguments.



Anchor

The team member in charge of *valuation*. A team captain (although not a team manager) who facilitates credible evaluations and provides a reasonable basis for decisions on whether to pivot or persevere.



The Problem-Solution Canvas

Exercise 2 (Start on your PSC)

Use the results from **Exercise 1** and the **PSC Template** in this exercise.

Start on the PSC for your project:

- Discuss and choose **Fulcrum** (Sections 4.1 – 4.5). Explain your choice.
- Based on your Fulcrum, broadly outline a **strategy** for your work on the problem and its solution.
- Begin with the **key cells** in the PSC based on your strategy. Keep it concise.

Building Blocks

Problem



We see a problem as a situation to be handled, where problems appear as manifestations.

Outer Environment



The problem is found in the outer environment. The design is adapted to this domain.

Manifestations



Manifestations are objects, actions, or events that reflect or embody the problem.

Capabilities



Capabilities are the features that are designed to address manifestations.

Inner Environment



The inner environment is what provides the capabilities.

Leverage



Leverage points allow for the most valuable capabilities.

Solution



Why the problem is resolved, why it is important, and why the result is appealing.

Evolvability



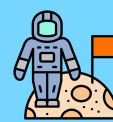
The capacity to adapt to outer environments or to add new leverage points to the inner environment.

Merit



The value offered by the capabilities. The limitations of the solution and why these limitations are acceptable.

Mission



The mission presents the project's tactical objective: to solve the problem.

Potential



The potential defines the project's strategic objective: to create strengths and opportunities.

Horizon



Where might the outcomes of this project lead us? How do they align with the future we aim to be a part of?

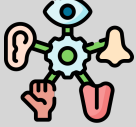
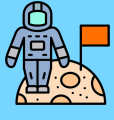
Illustration: ClusterDetect

	 PERCEIVE	 FORM	 CONSOLIDATE	 LEARN
 RATIO- NALE	 PROBLEM In crowded areas, safety could be compromised	 LEVERAGE Technology: AI [DFNet] Components: [Thermal camera] Information: Problem spots Human resources: [On-the-ground staff]	 SOLUTION PROSPECT ClusterDetect reduces congestion and disruption WARRANT Densely populated areas are common and dangerous BACKING The Contribution is effective and inexpensive	 HORIZON The project creates strengths and opportunities - The Problem is understood and generic - The Leverage is future-proof - The Prospect solves the Problem, - The Warrant suggests growth - The Backing suggests motivation to adopt
 STRATEGY	 OUTER ENVIRONMENT External service: [Access control] Implements: Venue information system Repository: Traffic information People: [Informants]	 INNER ENVIRONMENT Density monitor [Thermal?] Panic and disruption locator General modules for on-the-ground staff and visitors [Customized module for on-the-ground staff]	 EVOLVABILITY DIFFUSIBILITY ClusterDetect is highly specialized and has low diffusibility towards new markets ADOPTABILITY ClusterDetect is limited to the current market, but it can address new needs and thereby strengthen existing market positions	 POTENTIAL The design has strategic potential: - Outer Environment is generalized - Inner Environment is near-decomposable - The Qualifications are non-critical
 TACTICS	 MANIFESTATIONS Potential crowding Places of congestion Potential starts of panic Potentially disruptive actions	 CAPABILITIES Identify and handle crowds Identify and handle congestion Identify and handle panic Identify and handle disruption	 MERIT VALUE Locates impediments. Finds and handles crowds, congestion, panic, and disruption RESERVATION Requires calibration and clear sight REBUTTAL Many users will accept these limitations in a low-cost system	 MISSION ClusterDetect solves the Problem: - The Manifestations are essential - The Capabilities are efficient and effective - The Value offered ensures sustained use

Visual Inquiry Tools

- Visual inquiry tools support joint visual inquiry.
- They support reflection and collaboration among designers.
- They represent complex ideas as a whole, split into parts.
- Separating complex ideas into parts allows for overview and detailed inquiry.

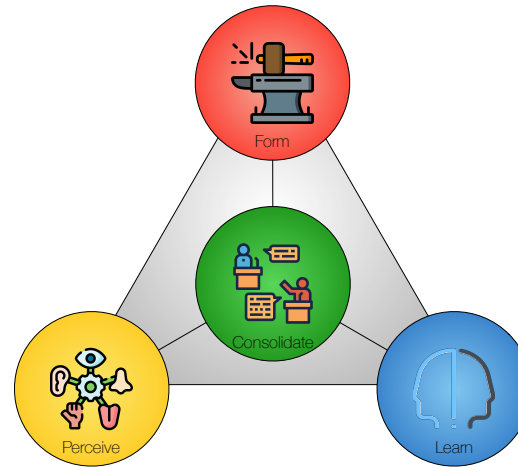
VIT: Problem-Solution Canvas

	 PERCEIVE	 FORM	 CONSOLIDATE	 LEARN
 RATIONALE	 PROBLEM	 LEVERAGE	 SOLUTION PROSPECT WARRANT BACKING	 HORIZON
 STRATEGY	 OUTER ENVIRONMENT	 INNER ENVIRONMENT	 EVOLVABILITY DIFFUSIBILITY ADOPTABILITY	 POTENTIAL
 TACTICS	 MANIFESTATIONS	 CAPABILITIES	 MERIT VALUE RESERVATION REBUTTAL	 MISSION

Fulcrums

- **Manifestation** Fulcrum
- **Capability** Fulcrum
- **Merit** Fulcrum
- **Mission** Fulcrum





3

Rationale & Problem

Exercise 3 (Problem Scenarios)

Use the results from [Exercise 2](#) and the [PSC Template](#) in this exercise.

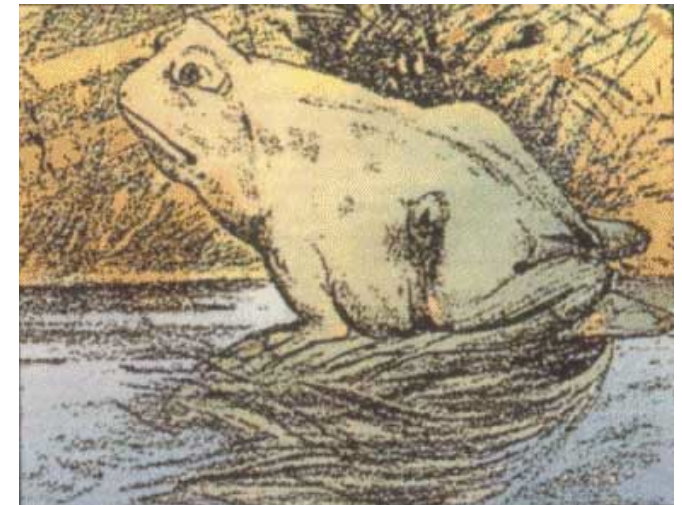
- Describe the [context type](#) of your project as defined in Table 6.1.
- Develop [axes](#) for Problem Scenarios as described in Section 6.3 (Tables 6.2, 6.3, 6.4, and 6.5). Choose two axes for the Problem Scenarios.
- Develop Problem Scenarios as illustrated in Figure 6.1. Explain the problem found in each [quadrant](#).
- [Choose](#) one problem to work with from now on. You can choose one quadrant or combine several if relevant. Express this problem in [one sentence](#) in the PSC problem cell (Section 3.1).
- Outline how this problem [manifests](#) itself in the PSC manifestations cell (Section 3.3).
- Outline the [Outer Environment](#) of your project and enter these elements in the Outer Environment cell (Section 3.2).

Situation

What is designated by the word "situation" is not a single object or event or set of objects and events. For we never experience nor form judgments about objects and events in isolation, but only in connection with a contextual whole. This latter is what is called a "situation."

This definition introduces a number of concepts in Deweyan pragmatism:

- Object
- Event
- Contextual whole
- Experience
- Judgment (appreciation)

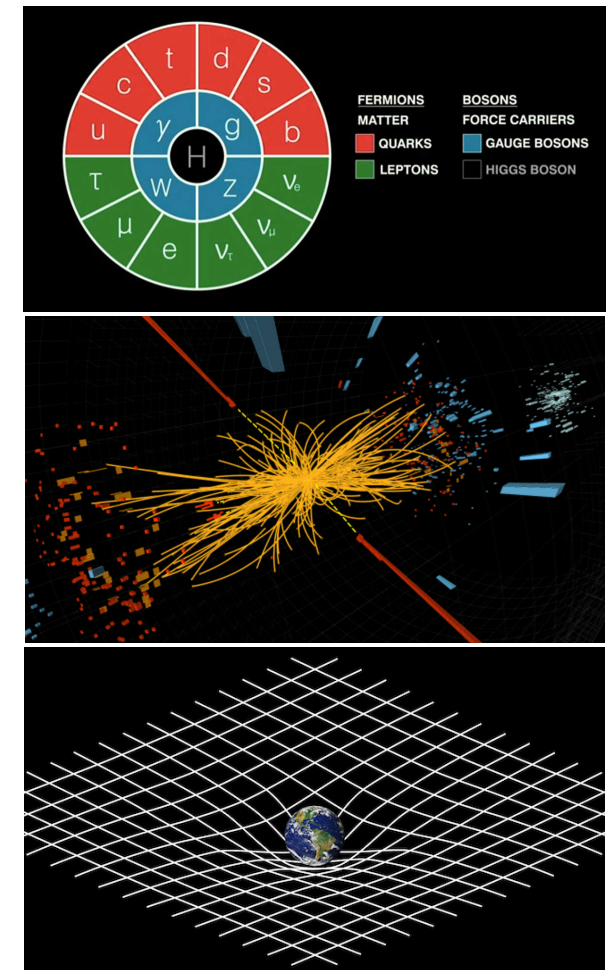


What is a Problem?

- *A problem represents the partial transformation by inquiry of a problematic situation into a determinate situation.*
 - *Situation:* Problems are contextual – conceived in a world of objects and events.
 - *Partial transformation:* Suggests a fleeting quality – problems may replace other problems as our understanding develops.
 - *Inquiry:* Problems are part of an intellectual effort – problems are perceptions that serve as instruments in this effort.

Existential vs. Ideational

- *Existential subject-matter* can interact with and modify other existential subject-matter
 - E.g. components, builds, objects and events.
- *Ideational subject-matter* can explain the situation and suggest what to do
 - E.g. planned components, objects and events, procedures, criteria, ideas.



Higgs Boson and Gravitational Waves exemplify ideational subject-matter turned existential

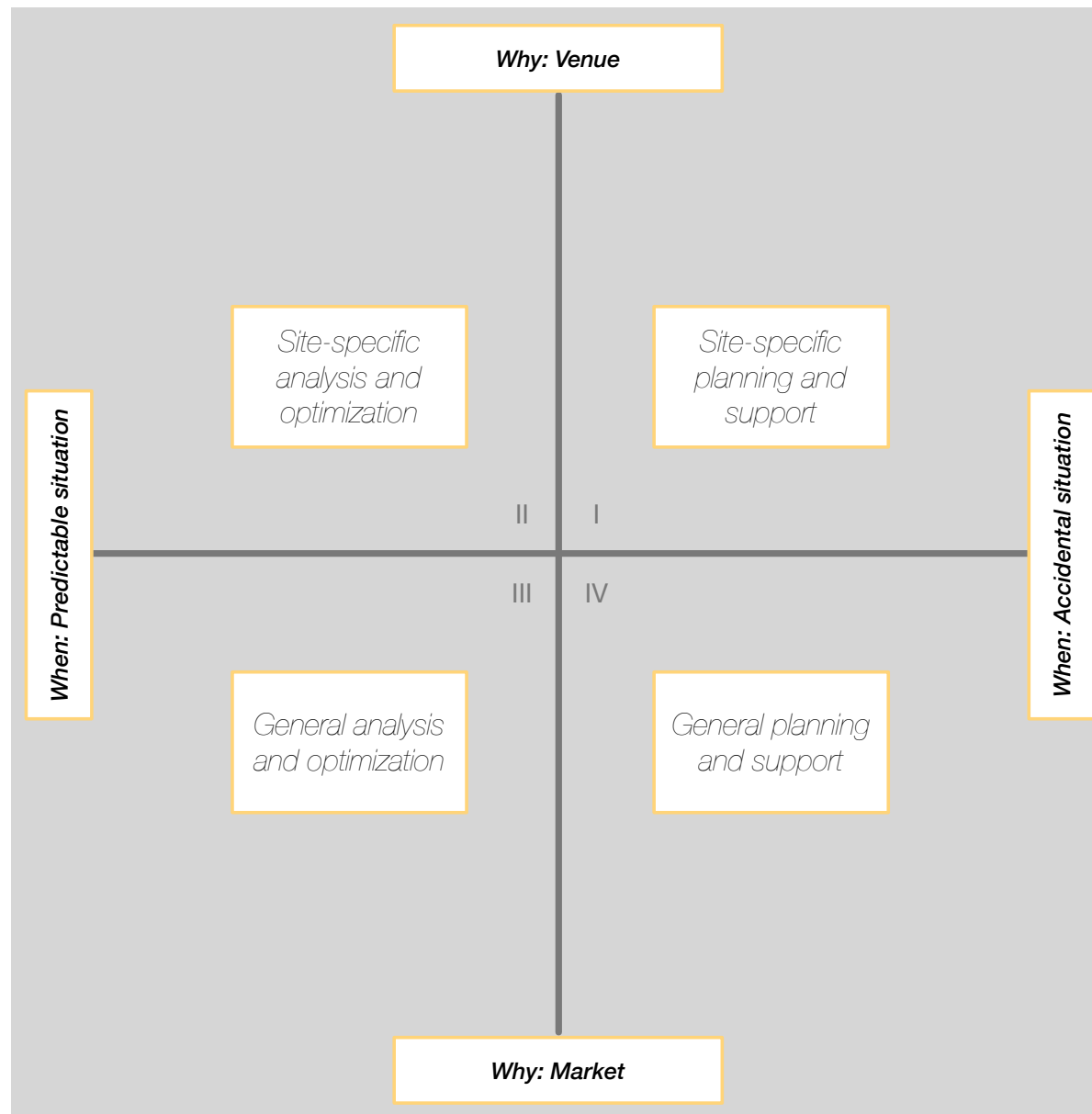
Four Context Types

CONTEXT	CHARACTERISTICS
<i>Simple</i>	The domain of <i>known knowns</i> . This is where there is a clear relationship between cause and effect. Where this relationship is evident to everyone, and where there is a right answer. In this domain we sense, <i>categorize</i> , and respond.
<i>Complicated</i>	The domain of <i>known unknowns</i> . This is where cause-and-effect relationships are discoverable but not obvious to everyone, and where there might be multiple right answers. In this domain we sense, <i>analyze</i> , and respond.
<i>Complex</i>	The domain of <i>unknown unknowns</i> . This is where we can't know a priori what will work, and where patterns emerge for handling this flux and unpredictability. In this domain, there are many competing ideas and no right answers. Here we <i>probe</i> , sense, and respond.
<i>Chaotic</i>	The domain of <i>unknowables</i> – the domain of Knightian uncertainties. This is where cause-and-effect relationships cannot be determined and where there is high turbulence, high tensions, and limited time. In this domain, we <i>act</i> , sense, and respond. We look for what works to transform from chaos to complexity and then identify emerging patterns.

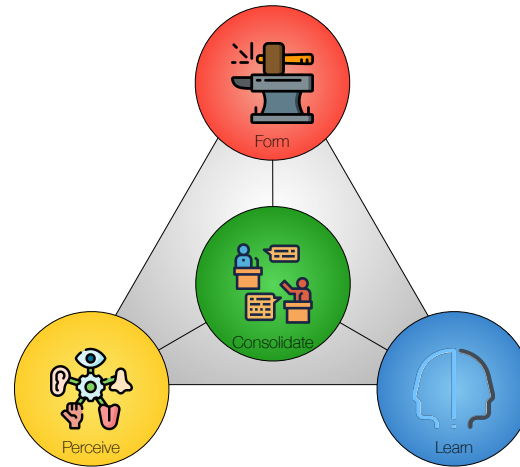
(Adapted from Snowden and Boone, 2007)

Problem Scenarios

ClusterDetect



Example based on one among many alternative axes



4

Leverage

Exercise 4 (Leverage Scenarios)

Use the results from [Exercise 3](#) in this exercise.

- Suggest [leverage points](#) for your project. Use the categories in Table 7.1 for inspiration, or develop your own.
- Evaluate and select [leverage points](#). Use [PCRT](#)-analysis as shown in Table 7.3 and/or Table 7.4.
- Develop axes for the [Leverage Scenarios](#) as described in Section 7.3 (Tables 7.5, 7.6, 7.7, and 7.8).
- Develop [Leverage Scenarios](#) as illustrated in Figure 7.1.
- Explain the [Leverage Scenarios](#) you find in each quadrant.
- Choose one scenario to work with from now on. You can choose one quadrant or combine several if relevant.
- Discuss the most important leverage points you plan to use in your project. Enter these leverage points in the [Leverage](#) cell (Section 3.6).
- Discuss the capabilities obtainable through these leverage points. Enter these capabilities in the [Capabilities](#) cell (Section 3.4).
- Discuss the Inner Environment of your design. Enter the key modules in the [Inner Environment](#) cell (Section 3.5).

Means – Material and Procedural

- *Means* are instruments used to attain an end – something useful in achieving a result.
- *Material means* – including observed data, facts and artifacts – combined serve to create a resolved situation.
- *Procedural means* – including operations, plans, requirements, and test cases – serve to determine how and if such a resolved situation is attained.

ClusterDetect material means: Number of people, legal limitations, sensors ...
ClusterDetect procedural means: PSC, experiments, spikes, prototyping, ...

Ideas

Ideas are anticipated consequences (forecasts) of what will happen when certain operations are executed under and with respect to observed conditions.

To Dewey, ideas are similar to hypotheses

What is Leverage?

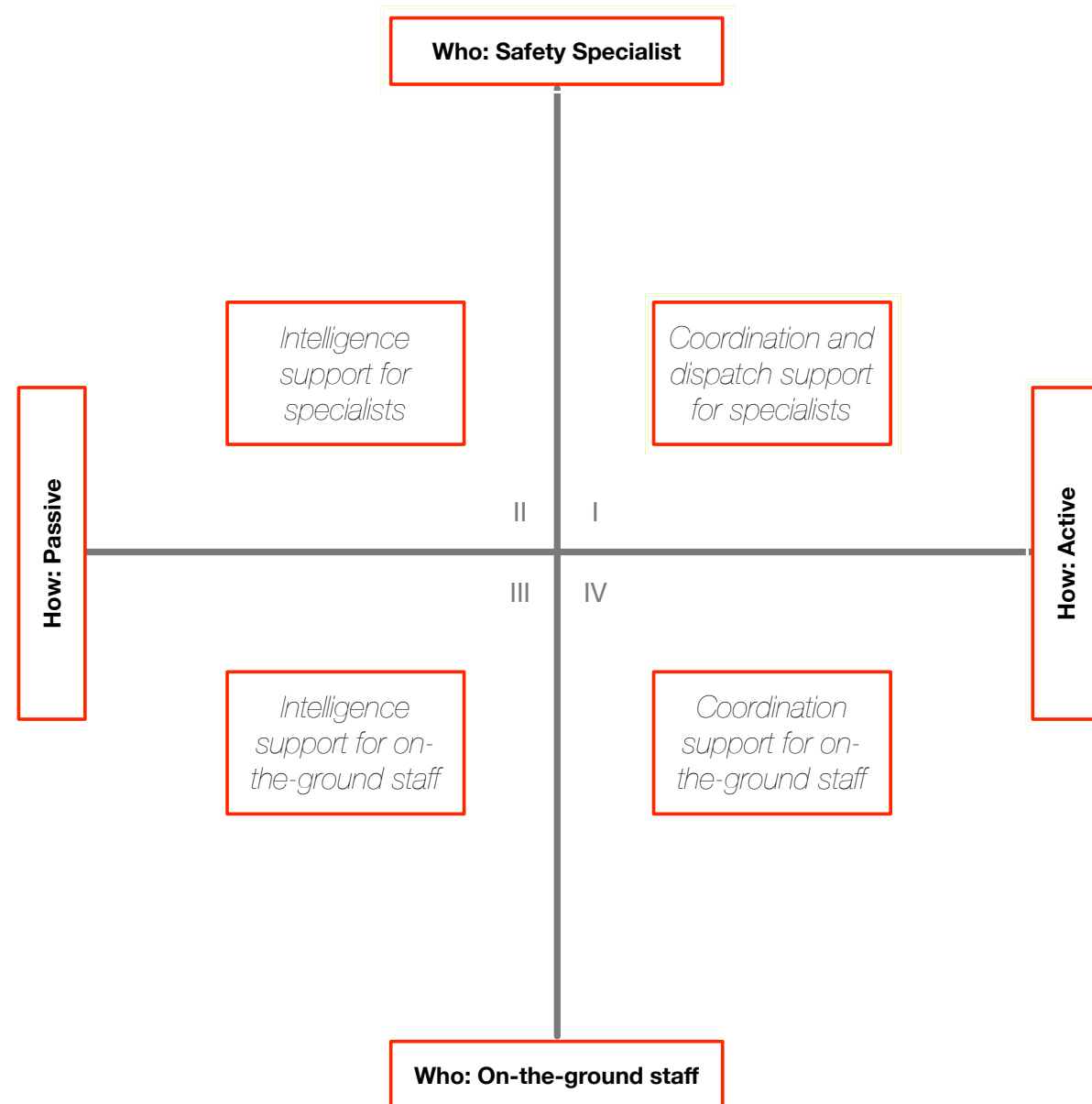
Leverage is short for leverage points – *technologies, components, information, or human resources* that could be keystones in an excellent solution.

A leverage point might lead to a *solution* with more substantial features, lower cost, greater flexibility, robustness, or any other *quality* to make a design stand out.

Proposed leverage points are *ideas*; they are forecasts pointing to possible effects on the project or the product. These forecasts are based on examinations of their *capacity* to help solve the problem.

Leverage Scenarios

ClusterDetect

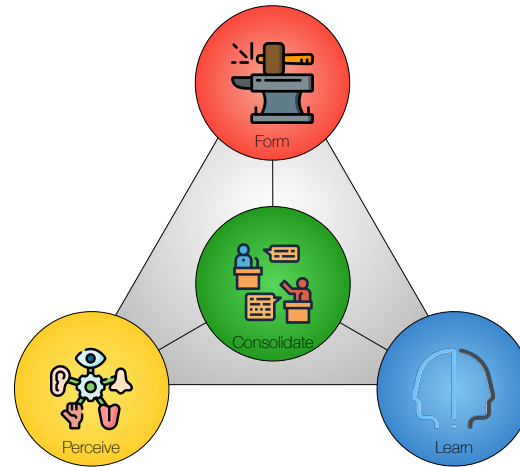


Example based on one among many alternative axes

Four Leverage Categories

CATEGORIES	CHARACTERISTICS
<i>Technology</i>	Scientific knowledge, machinery, and equipment on which a design might be based.
<i>Components</i>	Rare equipment, sensors, or actuators used actively in building a solution and accessed via crafted interfaces.
<i>Information</i>	Used for collecting, storing, disseminating, and/or aggregating information to be accessed via crafted interfaces.
<i>Human resources</i>	Specific users, categories of users, experts, authorities, social or professional networks, organizations, or others with particular roles or functions designed to build a solution. Cyberhuman systems are prominent examples of systems where internal people are leverage points.

Crafted interfaces indicate that items of these categories must be used as an integrated part of the design



5 Solution

Exercise 5 (Solution Scenarios)

Use the results from **Exercise 4** in this exercise.

- Develop **axes** for the Solution Scenarios as described in Section 8.3 (Tables 8.2 and 8.3).
- Develop **Solution Scenarios** as illustrated in Figure 8.1.
- Explain the **Solution Scenarios** you find in each quadrant.
- Choose one scenario to work with from now on. You can choose one quadrant or combine several if relevant.
- Discuss short-term **strengths** and **weaknesses** of your chosen scenario—how might this scenario create value and growth in the short run?
- Discuss longer-term **opportunities** and **threats** related to your chosen scenario—how might the future look if you use this scenario?

Ends and Ends-in-View

- An **end-in-view** is an idea of an **end** to be reached. It represents the **purpose** of taking a step.
- The end-in-view is the **anticipated** consequence of taking a step.
- An inquiry aims to resolve a problematic situation through **existential changes**.

Ends vs. Means

- “*End*” is a name for a series of acts taken collectively – like the term army.
- “*Means*” is a name for the same series taken distributively – like this soldier, that officer.
- To think of the end signifies to extend and enlarge our view of the act to be performed.
- It means to look at the next act in perspective, not permitting it to occupy the entire field of vision.
- The end thus re-appears as a series of “what nexts.”

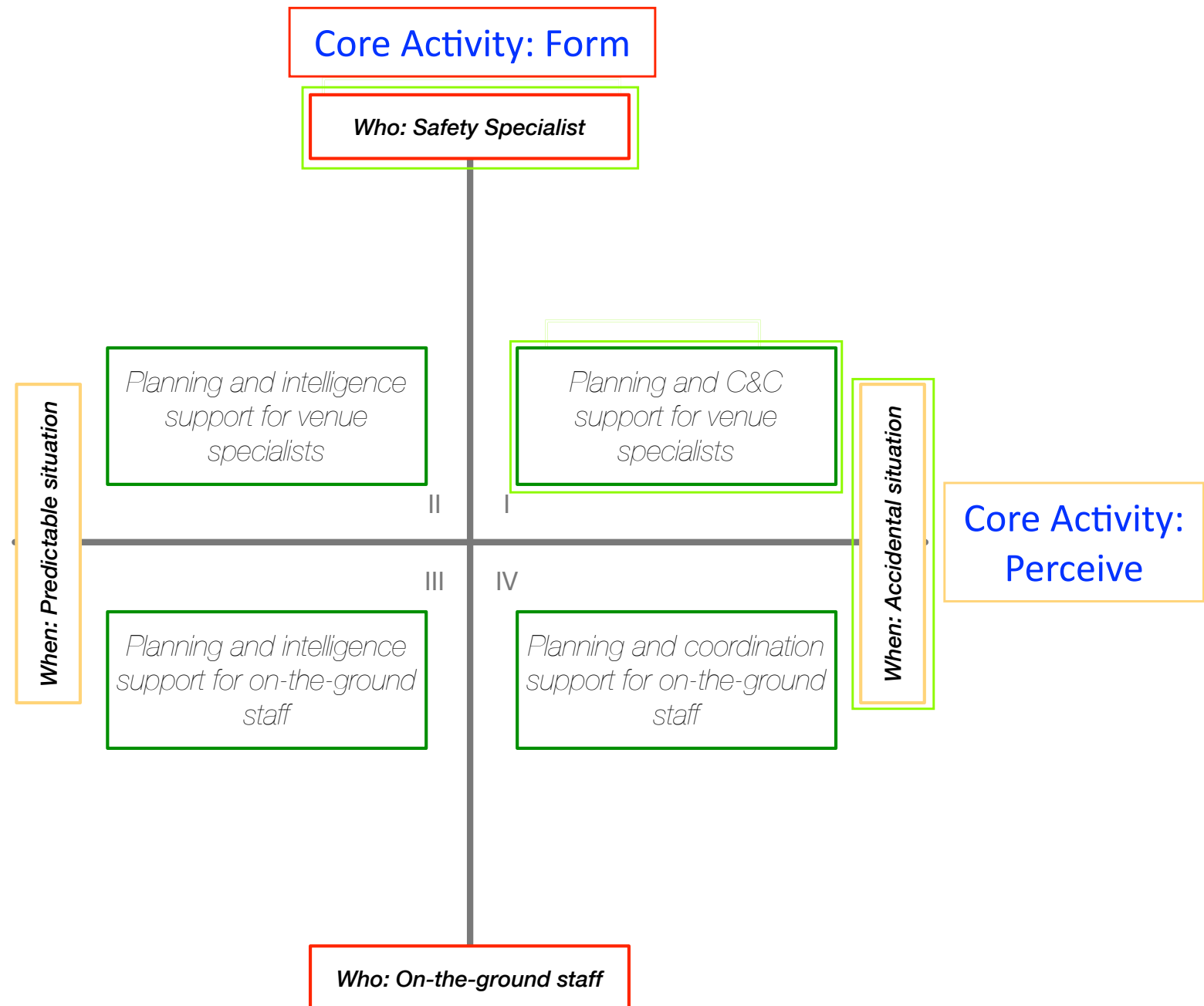
Prospect

- What we believe to be a solution if the products developed in the project are put to proper use.
- What Dewey would call our overall **end-in-view**.
- A prospect reflects an understanding of the problem and an adequate way to solve it.
- May span several iterations or sprints.
- A project may have the same prospect from start to end.

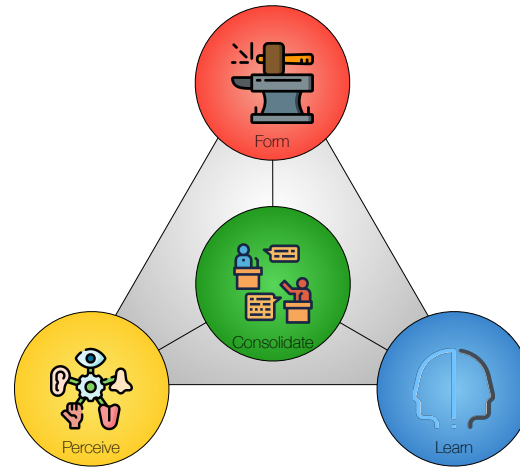
Prospect:

A mental picture of something to come.

Solution Scenarios—ClusterDetect



Example based on one among many alternative axes



6 Horizon

Exercise 6 (Horizon)

Use the results from **Exercise 5** in this exercise. This exercise is about the **project's horizon**.

- Will your horizon be affected by the **technology push**? What and how?
- Will your horizon be affected by **market pull**? What and how?
- Will changes in **innovation type** characterize your horizon? How and why?
- Formulate **horizon hypotheses** as outlined in Table 9.1. Make sure you consider the future as well as the present.
- Outline some **criteria** you might use to verify or falsify your **hypotheses**.

End-in-views in Essence

In Essence, we have two types of end-in-view:

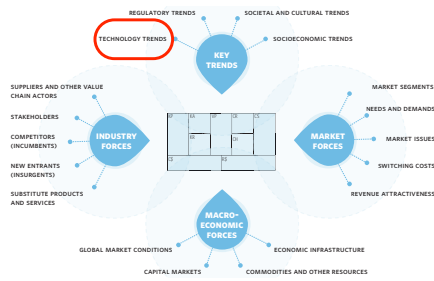
- *Mission end-in-view*, which pertains to the **current** project.
- *Horizon end-in-view*, which focuses on preparing for a **future** beyond the current project.

Both types can be **distant** and **subject to change**, as they may be elusive and outside our scope of knowledge.

Horizons in Essence

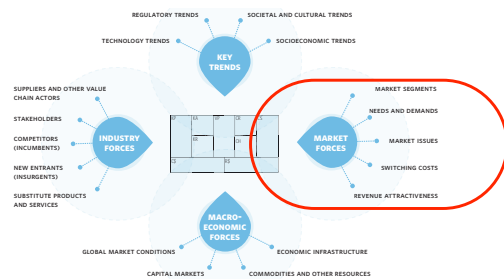
In Essence, we have three horizons:

- The first horizon (H1) represents business as usual—the way things are done now.
- The second horizon (H2) represents the shift and clash of different ideas about the future.
- Horizon Three (H3) is long-term and represents an emerging future pattern, including technologies, application areas, norms, values, and economics.



Technology Push

- Technology and science as drivers of change.
- Should we base future leverage on advancements in technology and science?



Market Pull

- Market features, users, economic developments, and more as **drivers of change**.
- Should we base **future leverage** on significant trends and changes in forces in the **market, industry, and economy**?

Purpose of Hypotheses

- We use **hypotheses** to ensure our design serves its **intended purpose** and meets our expected **standards** and **end-in-view**.
- A hypothesis is a **proposition** that gives us a basis for **reasoning** about something we need to **investigate** further.
- It is a starting point for **examining** whether the contribution **conforms** to the **expected** solution.
- Hypotheses are **verifiable** or **falsifiable** and predict specific outcomes.

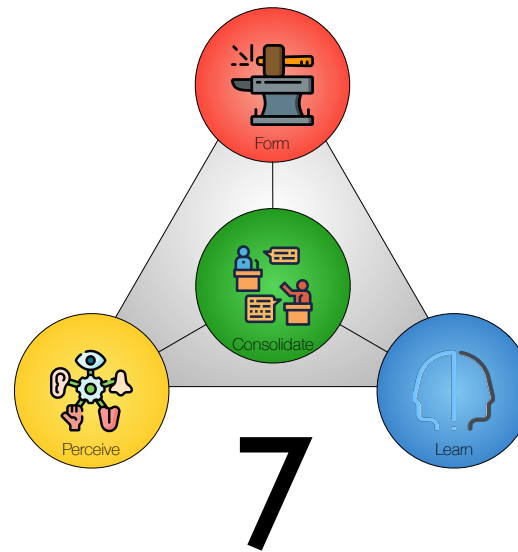
Horizon Hypotheses

Mission end-in-view

Horizon end-in-view

TIME	THE PRESENT	THE FUTURE
PROBLEM	The problem covers the issues faced by the stakeholders.	This problem will be significant in the future.
LEVERAGE	The leverage is vital for this project.	The leverage developed will help solve future challenges.
PROSPECT	The prospect solves the problem well.	This prospect is relevant to a category of future challenges.
WARRANT	The problem is worth solving.	This solution will help address future problems worth solving.
BACKING	The solution is effective and competitive.	This project will facilitate effective and competitive future solutions.

Table 9.1: Horizon hypotheses



Strategy & Outer Environment

Exercise 7 (Strategy & Outer Environment)

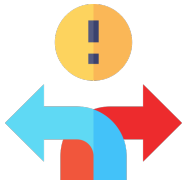
This exercise is about **VUCA** strategies for the **Outer Environment**.

- Discuss VUCA **challenges** for your project (see Table 10.1). Which of these challenges are most likely to affect your project?
- Discuss VUCA **strategies** for your Outer Environment (see Table 11.2). Are they useful? How? Suggest **alternative** strategies if they are not useful.
- How could you use these strategies **actively** in your project?

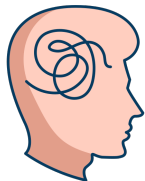
VUCA



- *Volatility* – i.e., conditions that change rapidly or unpredictably.



- *Uncertainty* – i.e., needs and solutions that are unsettled.



- *Complexity* – i.e., external or internal parts that form intricate meshes.



- *Ambiguity* – i.e., causes and effects that are mixed up.

Strategic Challenges for Design

		KNOWLEDGE ABOUT THE SITUATION	
ABILITY TO PREDICT THE RESULTS OF ACTIONS	HIGH	COMPLEXITY  External or internal parts that form intricate meshes. The situation or contribution has many interconnected parts and variables. Some information may be available, but the volume can be overwhelming.	VOLATILITY  Conditions that change rapidly or unpredictably. The situation or contribution is unstable or may change unexpectedly, but is not necessarily hard to understand. Knowledge about the change is often available.
		AMBIGUITY  Causes and effects that are mixed up. Causal relationships are completely unclear. The logic of the situation changes with our actions, and stakeholders might be unsure of their preferences.	UNCERTAINTY  Needs and solutions that are unsettled. Basic causes and effects are known, but there is a lack of knowledge about possible changes. The future is, therefore, unpredictable despite available information.
	LOW		
		Low	HIGH

Inspired by (Bennett and Lemoine, 2014)

The Outer Environment

Our environments will not be *given* but *made*, and increasingly, they will be made independently of us by somebody else in the shape of new technologies, systems, services, sources of information, global or national standards, etc.

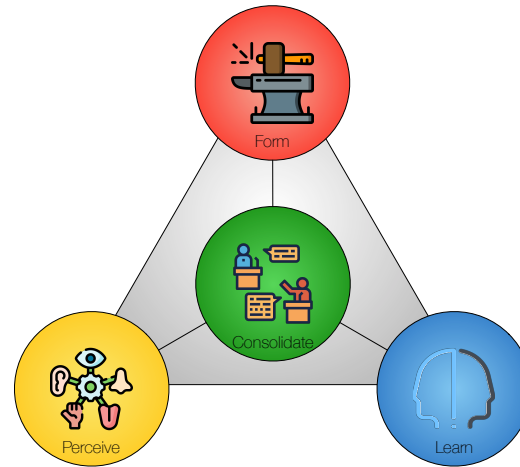
- In many cases, our environments will be *formed* or *constructed* by the actions of *others*.
- In this sense, our environment is *unknown* from the outset, yet we will need to start the project based on *assumptions* about it.

Four Outer Environment Categories

CATEGORIES	CHARACTERISTICS
<i>External services</i>	Services that are used without modification and reduce what is left for us to deal with.
<i>External implements</i>	Equipment, sensors, or actuators available in the outer environment and accessed via trivial interfaces.
<i>External repositories</i>	Repositories for collecting, storing, disseminating, and/or aggregating information via trivial interfaces.
<i>External people</i>	Generally available people, professionals, stakeholders, authorities, networks, organizations, etc.

VUCA Strategies for Outer Environment

		KNOWLEDGE ABOUT THE SITUATION	
ABILITY TO PRE-DICT THE RESULTS OF ACTIONS	HIGH	COMPLEXITY  External or internal parts that form intricate meshes. STRATEGIES • Fewer manifestations • Fewer user categories • Fewer dependencies	VOLATILITY  Conditions that change rapidly or unpredictably. STRATEGIES • Stockpile resources. Include force majeure clauses • Act, sense, and respond
	LOW	AMBIGUITY  Causes and effects that are mixed up. STRATEGIES • Probe, sense, and respond • Look for patterns to suggest trends and sense-making	UNCERTAINTY  Needs and solutions that are unsettled. STRATEGIES • Prepare contingency plans • Isolate dependencies
		LOW	HIGH



8

Inner Environment

Exercise 8 (Inner Environment)

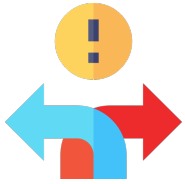
This exercise is about **near-decomposability** and **VUCA** strategies for the **Inner Environment**.

- The introduction part of chapter 12 (pages 140-141) discusses the use of **conceptual models** for developing and revising design ideas. Which conceptual models do you use to design the architecture and outline primary components in your project?
- Are any of these models **helpful** for modeling near-decomposable designs, as discussed in Section 12.2?
- Discuss VUCA **strategies** for your Inner Environment (see Table 12.1). Are any of them relevant to your project? How? Suggest **alternative** strategies if they are not useful.
- How could you use these strategies **actively** in your project?

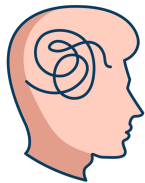
VUCA



- *Volatility* – i.e., conditions that change rapidly or unpredictably.



- *Uncertainty* – i.e., needs and solutions that are unsettled.



- *Complexity* – i.e., external or internal parts that form intricate meshes.



- *Ambiguity* – i.e., causes and effects that are mixed up.

Strategic Challenges for Design

		KNOWLEDGE ABOUT THE SITUATION	
ABILITY TO PREDICT THE RESULTS OF ACTIONS	HIGH	COMPLEXITY  External or internal parts that form intricate meshes. The situation or contribution has many interconnected parts and variables. Some information may be available, but the volume can be overwhelming.	VOLATILITY  Conditions that change rapidly or unpredictably. The situation or contribution is unstable or may change unexpectedly, but is not necessarily hard to understand. Knowledge about the change is often available.
		AMBIGUITY  Causes and effects that are mixed up. Causal relationships are completely unclear. The logic of the situation changes with our actions, and stakeholders might be unsure of their preferences.	UNCERTAINTY  Needs and solutions that are unsettled. Basic causes and effects are known, but there is a lack of knowledge about possible changes. The future is, therefore, unpredictable despite available information.
	LOW		
		Low	HIGH

Inspired by (Bennett and Lemoine, 2014)

The Inner Environment

The inner environment is our contribution – the substance and organization of our design and implementation.

This definition is based on Simon's definition: *The substance and organization of the artifact itself* (Simon, 1996, p. 6).

- The word artifact is Simon's term for what we call *design*.
- The artifact is the *meeting point* between the outer and inner Environment.

In Essence, the inner environment is the *architecture* and the *primary components* used to exploit the *leverage* and implement the desired *capabilities*.

VUCA & Leverage Points

CATEGORIES	CHARACTERISTICS
<i>Technology</i>	Scientific knowledge, machinery, and equipment on which a design might be based.
<i>Components</i>	Rare equipment, sensors, or actuators used actively in building a solution and accessed via crafted interfaces.
<i>Information</i>	Used for collecting, storing, disseminating, and/or aggregating information to be accessed via crafted interfaces.
<i>Human resources</i>	Specific users, categories of users, experts, authorities, social or professional networks, organizations, or others with particular roles or functions designed to build a solution. Cyberhuman systems are prominent examples of systems where internal people are leverage points.

Four leverage categories (chapter 5).

Leverage points might be affected by VUCA—
this could reduce (or increase) survivability

VUCA Strategies for Inner Environment

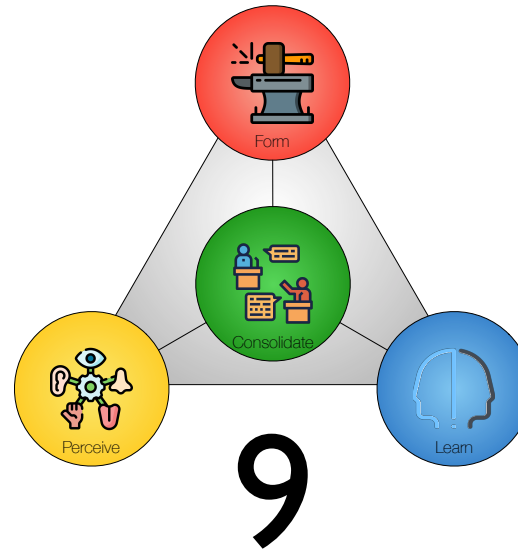
		KNOWLEDGE ABOUT THE SITUATION	
		Low	High
ABILITY TO PRE-DICT THE RESULTS OF ACTIONS	HIGH	COMPLEXITY External or internal parts that form intricate meshes.  STRATEGIES • Modularize • Divide & conquer	VOLATILITY Conditions that change rapidly unpredictably.  STRATEGIES • Build generic capabilities • Use microservice architecture
	LOW	AMBIGUITY Causes and effects that are mixed up.  STRATEGIES • Experimentation (MVP) • Problem-solving as natural selection	UNCERTAINTY Needs and solutions that are unsettled.  STRATEGIES • Information hiding (Parnas) • Isolate uncertainties in modules

V: *Generic* means reusable, lego parts. Make sure parts can contribute to other purposes

U: *Information hiding*. Make sure critical parts can be changed without side effects

C: *Modularize*: Simplify through abstraction. *Divide & conquer*: Pivot, zoom-in, simplify through reduction

A: *MVP*: try things out, select promising contributions. Search for capabilities that are valued. Test



Evolvability and General Innovation Theory

Exercise 9 (Evolvability & General Innovation Theory)

This exercise concerns the **evolvability** and **innovation characteristics** of your project.

- Describe and illustrate **diffusibility** in relation to your project.
- Describe and illustrate **adoptability** in relation to your project.
- Evaluate your project using one or more of the **evaluation types** presented on the slides.
- Describe your project using the terms **invention**, **innovation**, **exploitation**, and **diffusion**.
- Describe the **novelty**, **radicality**, and **timing** of your project.

The Spider Plant Metaphor



Diffusibility

We use the term **diffusibility** to describe the first part of **evolvability**: entering new environments—migrating across niches—and generating potential **variations** in search of new leverage points and application domains.

- **Variation** — creating probes (runners)
- **Selection** — choosing leverage points and domains that could become future bets on horizon H3, and using every opportunity to specialize probes toward those future bets.
- **Retention** — keeping strategic probes alive long enough to mature and demonstrate their value.

Diffusibility and Fulcrums

- *Manifestation fulcrums*—if similar manifestations appear in many other contexts, there is potential for wider diffusion.
- *Capability fulcrums*—if similar capabilities support many different tasks, they could have broader use.
- *Merit fulcrums*—innovative ways to generate value and drive growth if we create value for users willing to pay for it.
- *Mission fulcrums*—overcoming new challenges or surpassing existing limits. Whether this *invention* becomes an *innovation* depends on its broader use and its impact.

Adoptability

We use the term **adoptability** to describe the ability to **adapt to** and **fit into** new conditions. Adoptability refers to the **likelihood** that an innovation can be integrated into new domains.

- Adoptability can be **enhanced by design**: modular solutions enable partial adoption or adaptation into other application areas.
- **Retention** leads to the mutual selection of **leverage points** and **application domains** that complement each other.

Team Archetypes

		ADOPTABILITY	
DIFFUSIBILITY	HIGH	HIGH ROLLERS Many probes are used, but with a low survival rate. Very experimental but lacking clear direction. • Focus on trying out options • Losses occur more often than gains	PATHFINDERS Able to navigate in unknown territories. Good at probing and good at landing or scaling. • Focus on transformation • Optimal stance for unpredictable times
	LOW	RECREATORS Neither exploring new options nor demonstrating the ability to create new leverage points or markets. • Focus on conserving strengths • High-risk posture during times of disruption	STABLE PERFORMERS Able to deliver well if the direction is known. Less suited for exploring H ₃ options. • Focus on delivering • Sensitive to disruptions in markets and leverage points
		Low	
			HIGH

Table 13.1: Team Archetypes for Problem-solving and Innovation

Basic Terms

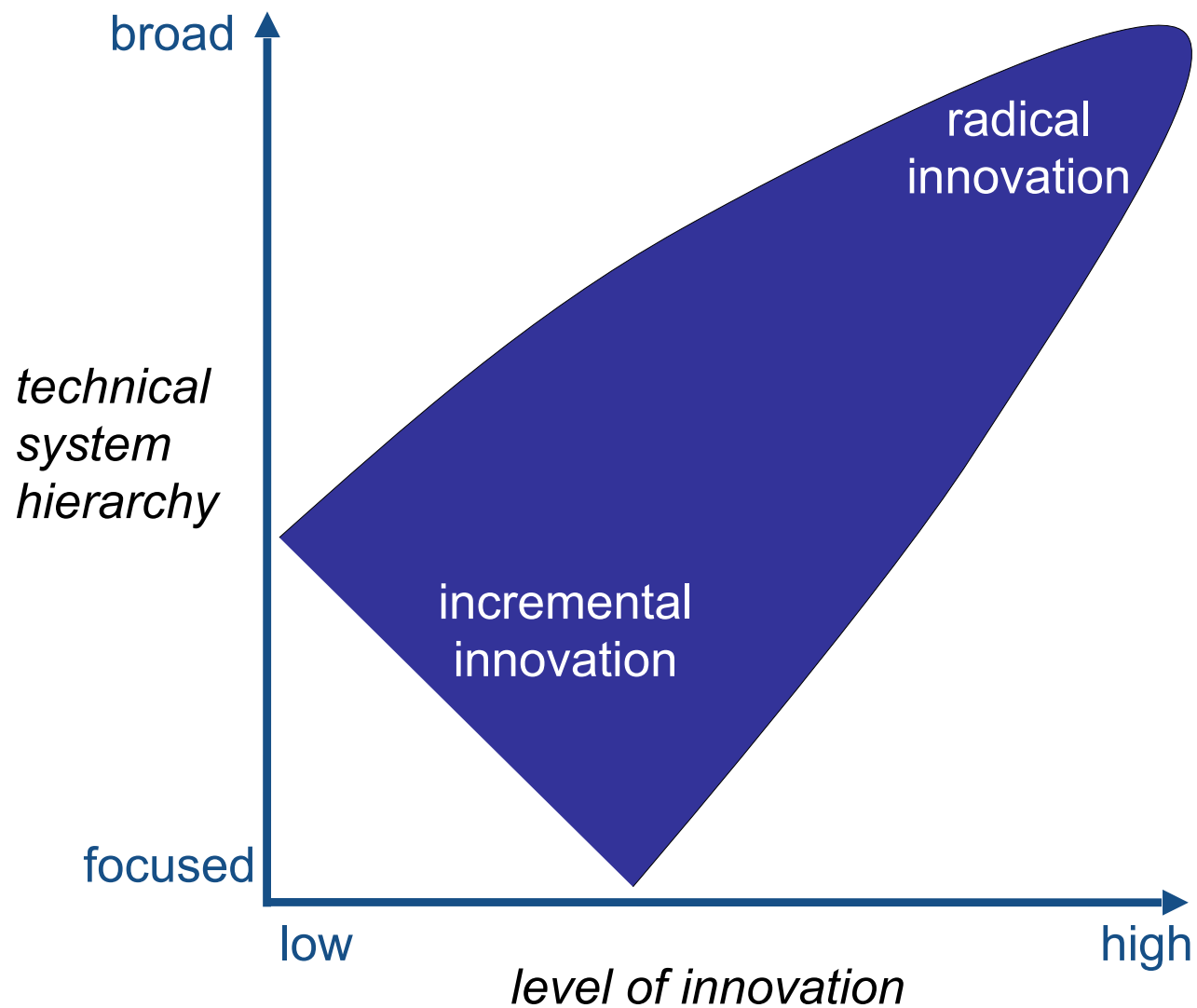
- *Creativity* – a state of mind which leads to inventive or innovative thinking.
- *Invention* - e.g., a new algorithm, new use of data, or program.
- *Innovation* - creative act and invention carried into wider use, leading to substantial kinds of change. Innovation implies the successful exploitation of new ideas.

Push and Pull

We may coarsely divide factors of economic or societal change into technology-push and demand-pull:

- *Technology-push*
Changes linked to the emergence and diffusion of new technologies: Digitalization, Artificial Intelligence, Internet of Things, Industry 4.0, smart sensors, etc.
- *Demand-pull*
When new solutions answer problems or needs: The seventeen UN Sustainable Development Goals (SDGs), climate change, pandemics, globalization, etc.

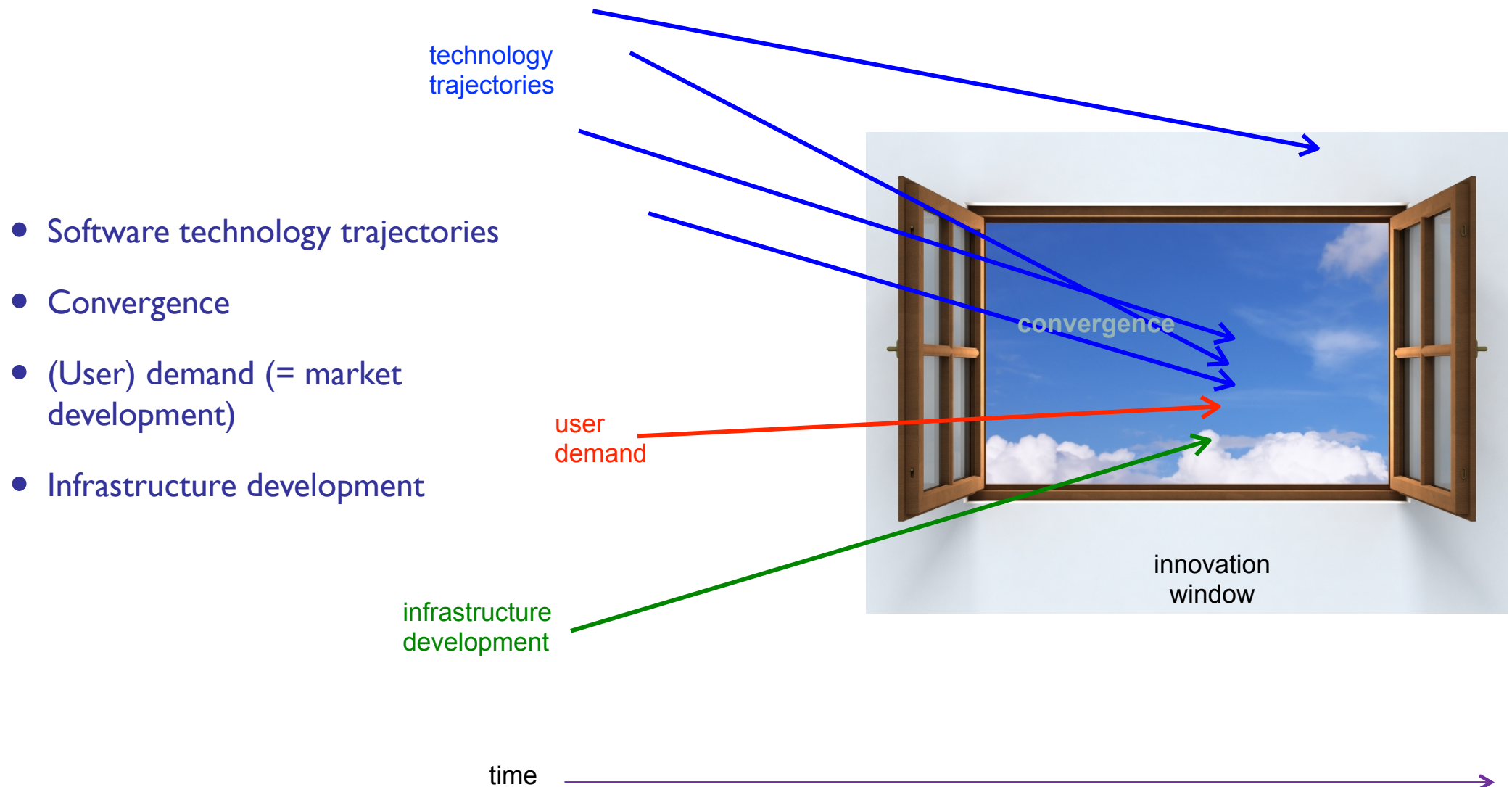
Incremental and Radical Innovation

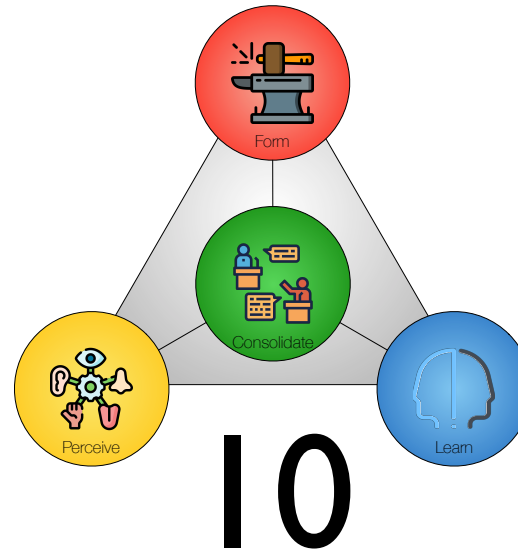


(software) innovation timing

- Trajectory.
- Digitalization.
- Convergence.
- Infrastructure.
- Innovation windows.

The Innovation Window





Tactics, Manifestations, Capabilities

Exercise 10 (Tactics, Manifestations & Capabilities)

This exercise concerns the **tactics**, **manifestations**, and **capabilities** of your project.

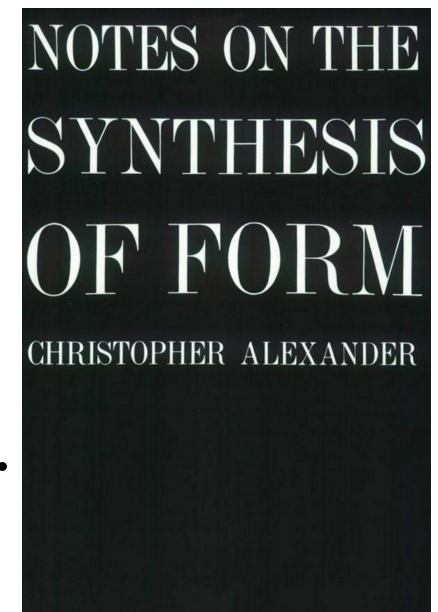
- Will you characterize your approach as **push** or **pull**?
- Describe concrete and abstract **manifestations**.
- Describe concrete and abstract **capabilities**.
- Will your abstractions improve **evolvability** and make your design **future-proof**? How?

Problem and solution formulation are linked

- Moving between the concrete and the abstract.
- Moving between the problem side and the solution side.
- Moving between the current situation and future options.

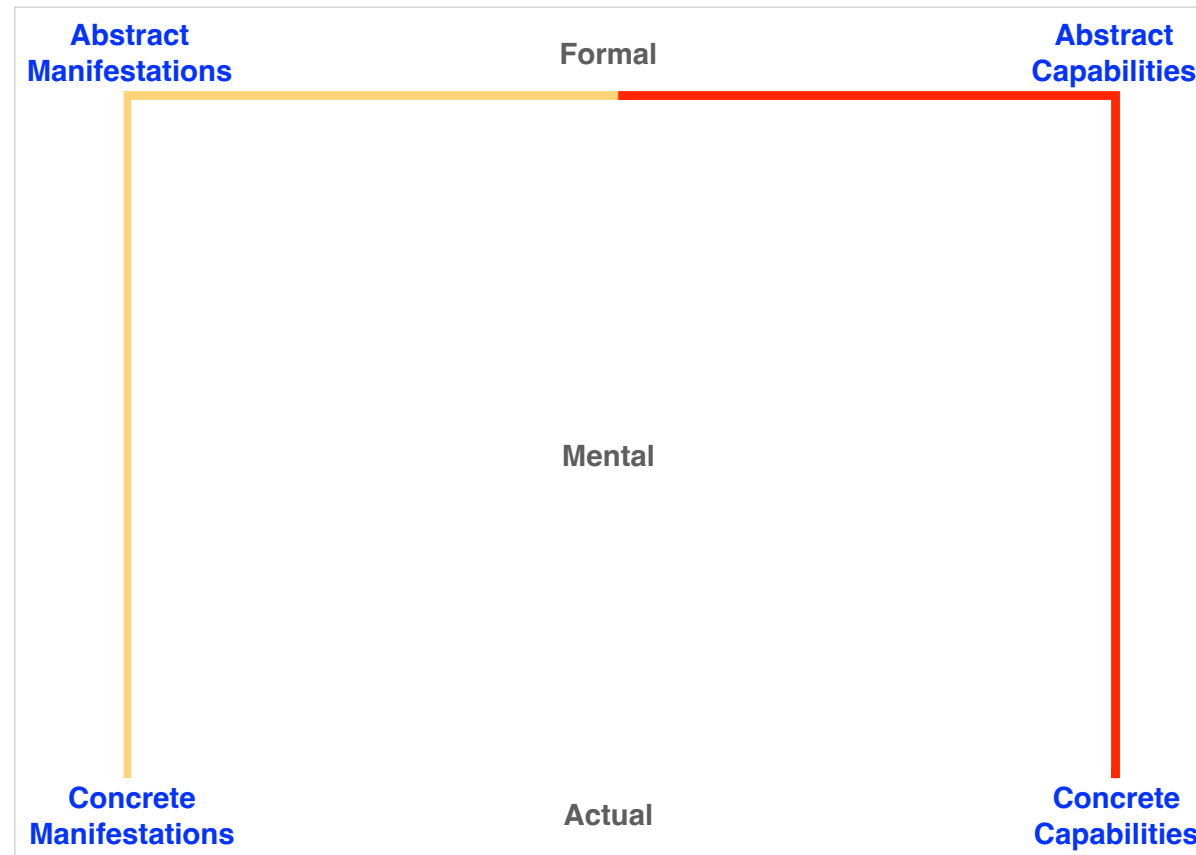
The Unselfconscious, the Selfconscious, and the Qualified

- The **unselfconscious** “designer” works only at the **level of the actual**. The problem-solver simply reacts to misfits by changing them.
- The **selfconscious** designer works also at the **mental level**. Design is shaped intuitively by a **conceptual** interaction between a mental picture of the context and a repertoire of ideas.
- The **qualified** designer also works on the **formal level**. Design is shaped by selecting **patterns** based on their pros and cons.



First out 1964

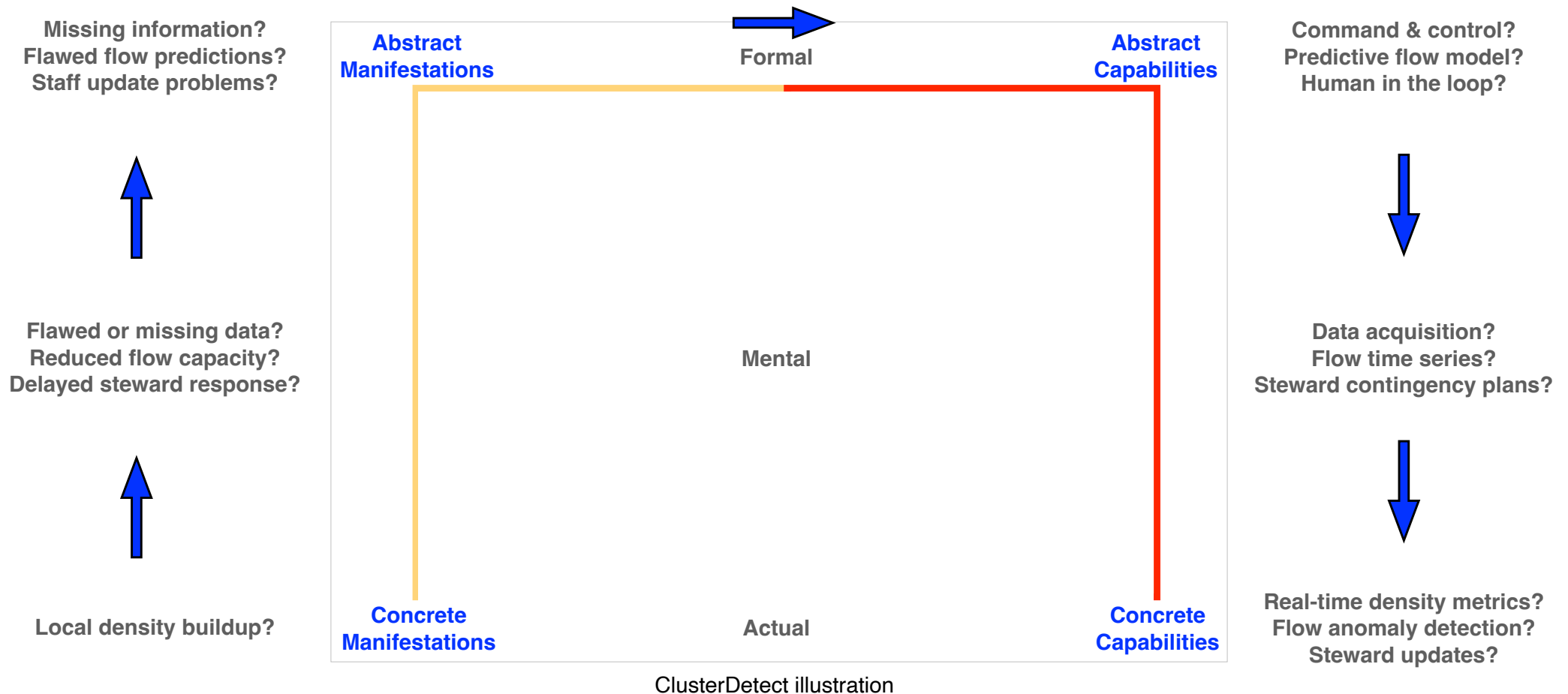
The **abstract** and the **concrete** are complementary



Inquiry is a movement between the **situation**, the **problem**, and the **solution**.
Push or Pull?—Inquiry may begin on either the problem or the solution side.

Concrete vs. Abstract

– Demand Pull

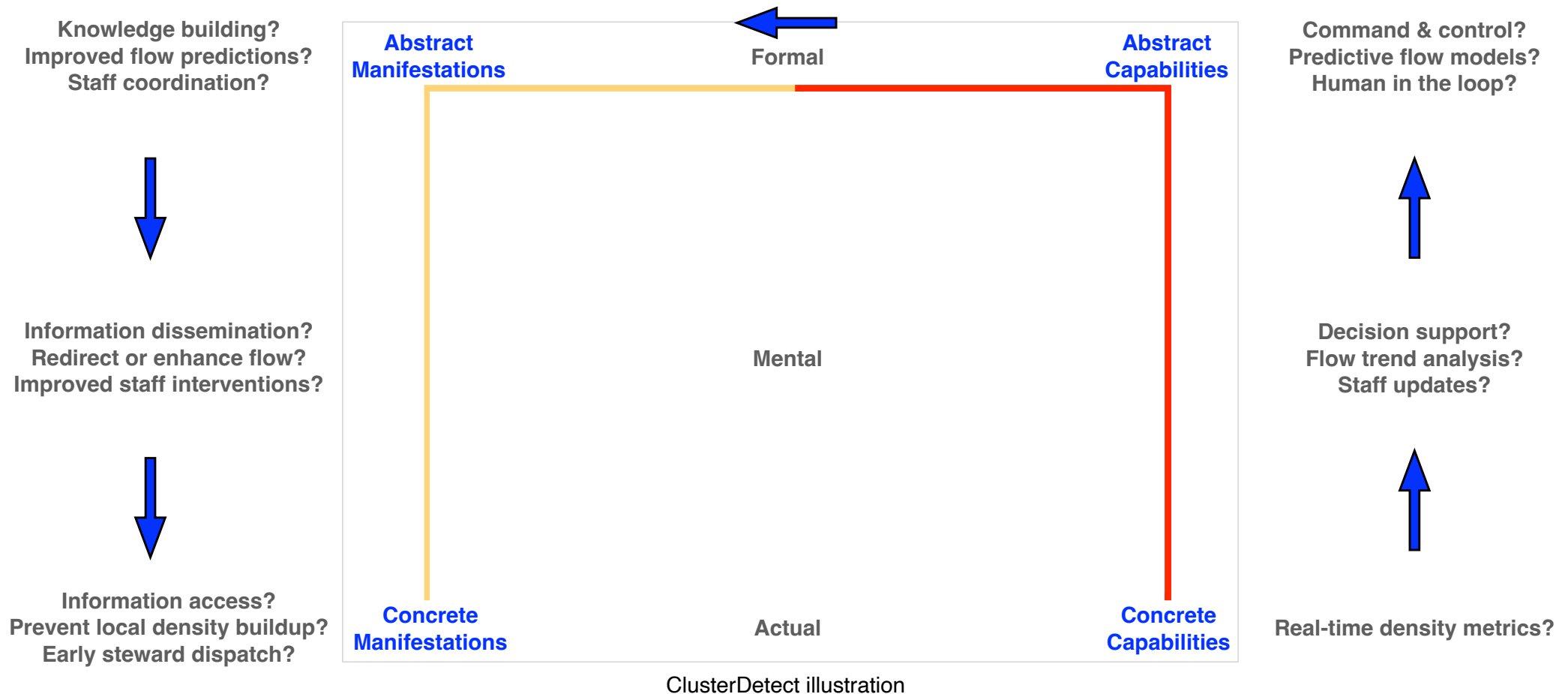


Pull—from problem toward solution. Making decisions on the way.

A problematic situation is noticed first. Designers identify manifestations, formulate the problem, and then develop a solution.

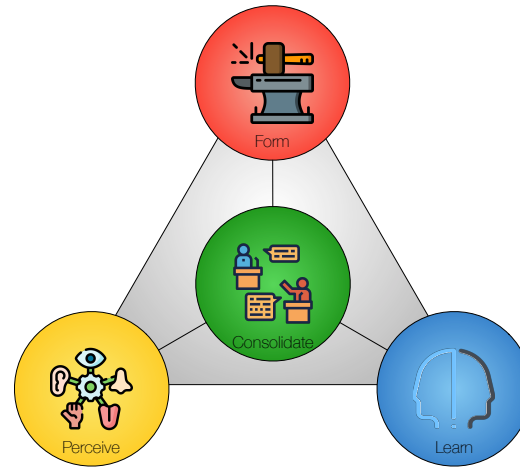
Concrete vs. Abstract

– Technology Push



Push—from solution toward problem

A technological possibility appears first. Designers explore what kinds of problems it may address, under what conditions, and for whom. What problematic situations could this help transform?



II

Merit

Exercise 11 (Merit)

This exercise concerns **functional** and **non-functional** requirements, as well as strategies for improving **robustness** and managing **dependencies**.

- Discuss the **functional** completeness of your design and state *reservations*, *rebuttals*, and *values* accordingly.
- Discuss the **robustness** of your design and assess if any of the strategies in section 18.2 would be helpful.
- Outline **dependencies** in your design and assess the utility of the strategies in section 18.3.
- Use **ETVX** (section 18.4) to model **robustness** and **dependencies**.

Functional and Non-functional Requirements

- Functional requirements are typically first-order concerns (part of PSC):
 - Is the solution functionally complete? Can we accept limitations?
- Non-functional requirements are typically second-order concerns (supplement to PSC):
 - Will it operate effectively?
 - Is it vulnerable to significant risks?

Stakeholders usually consider function primary—what they pay for.

Functional Merits

Functional requirements: Is the solution functionally complete? Can we accept limitations?

- **Value**—the functional value of the capabilities.
- **Qualifications**—a qualification is a statement or assertion that makes another less absolute.
- **Reservation**—limitations in the solution (why the design might not support a complete solution to the problem).
- **Rebuttal**—why limitations to the solution would be acceptable anyway.

Non-functional Merits

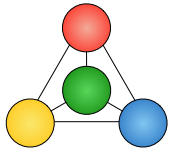
- Non-functional features may help prepare for unforeseen situations.
- Non-functional features may also introduce vulnerabilities and dependencies into our design.
- **Robustness**—what enables a system to withstand adverse or unexpected conditions.
- **Dependencies**—vulnerabilities caused by factors outside our immediate control.

Robustness Defined

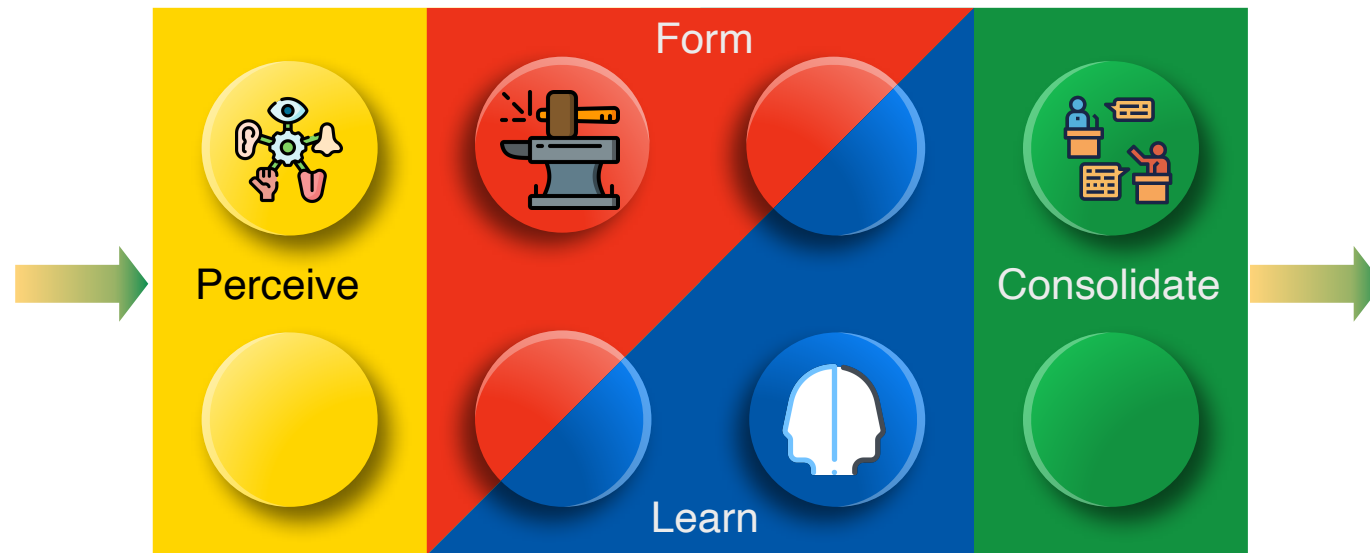
- Robustness means **insensitivity** to damages or the **ability to withstand** adverse and unforeseen events and consequences of human error without damage (SEBoK, 2024).
- In Essence, we will slightly expand this definition to encompass the **solution** we deliver and our development **strategies**.
- We do this to **prevent or mitigate risks** and to enhance the **capacity to adapt** to future technological changes, available services, working conditions, and more.

Dependencies

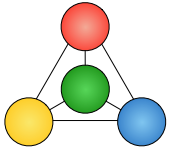
- Dependencies are **vulnerabilities** that arise from factors beyond our immediate control.
- These factors may stem from both the **outer** and **inner** environments.
- **External services** or **internal leverage points** on which our design depends are potential vulnerabilities:
 - When they work, they may help us build stronger solutions than we could on our own (for example, security services).
 - Ironically, potential **disruptions** of such could render our solutions **crippled** or **useless**.



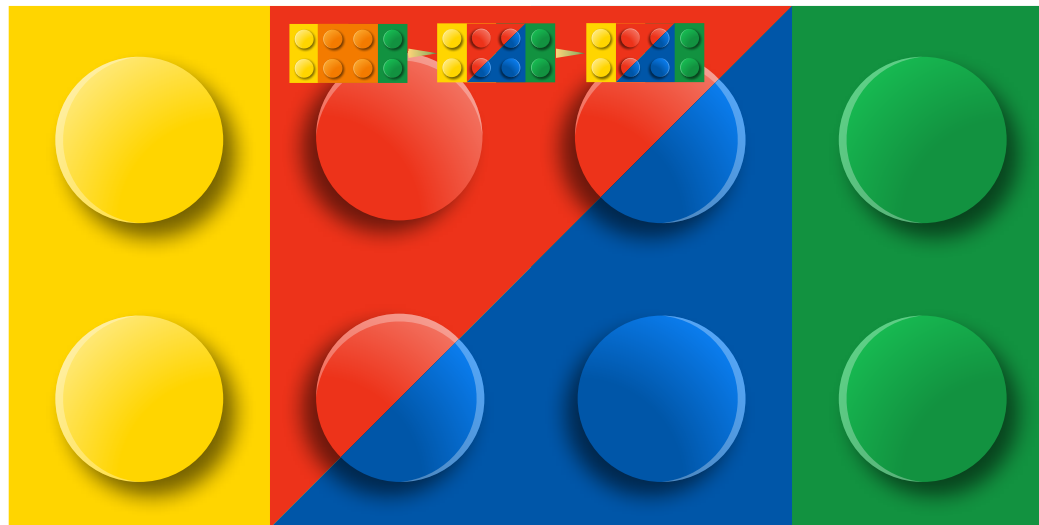
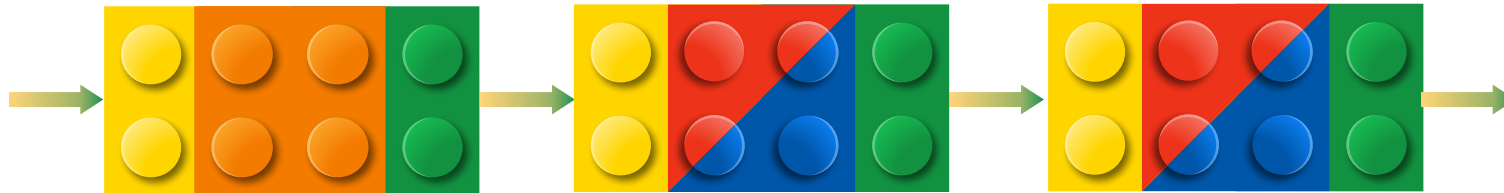
Essence adaptation of the ETVX model

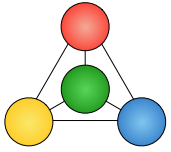


- **Entry** Criteria – criteria that should be satisfied before beginning the task.
- **Task** – descriptions of what is to be accomplished.
- **Validation** – procedures to ensure the quality of what is accomplished by the task.
- **eXit** – criteria that should be satisfied for the activity to terminate.

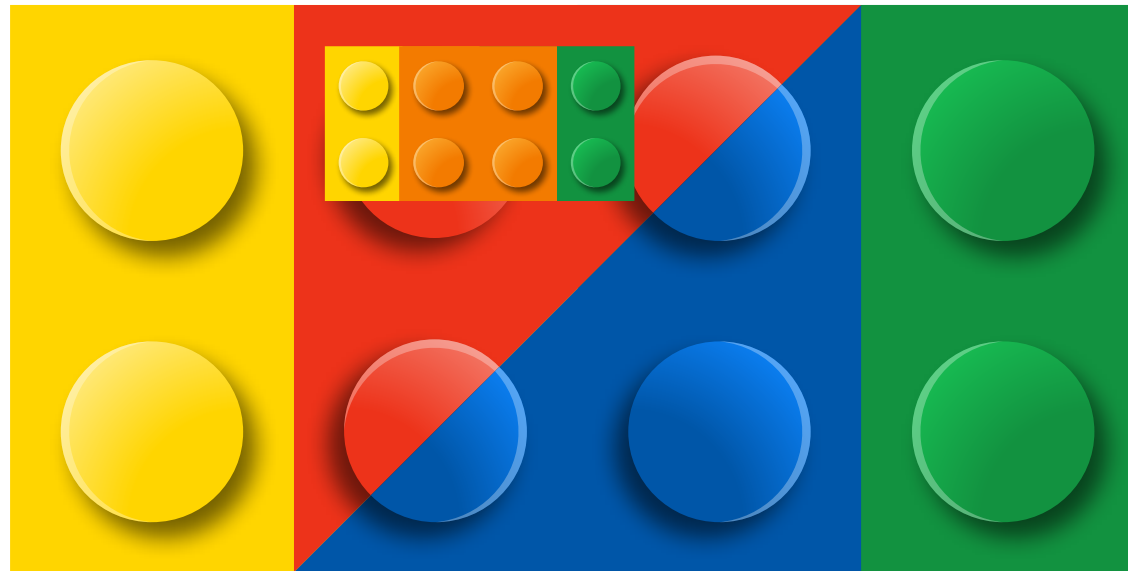


Cascading Cells

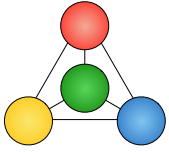




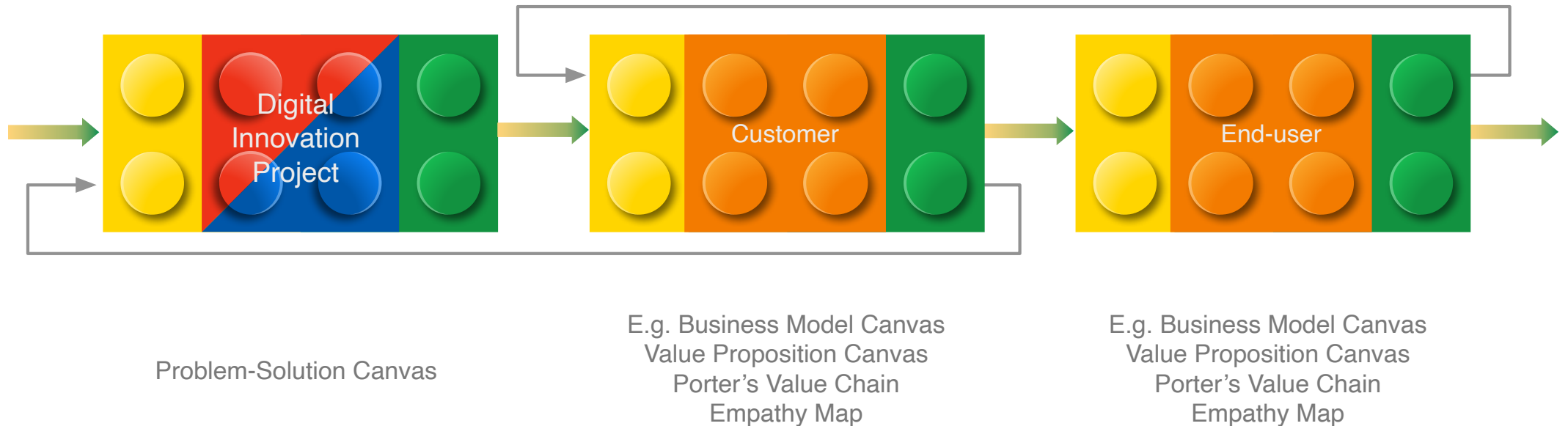
Abstraction Layers

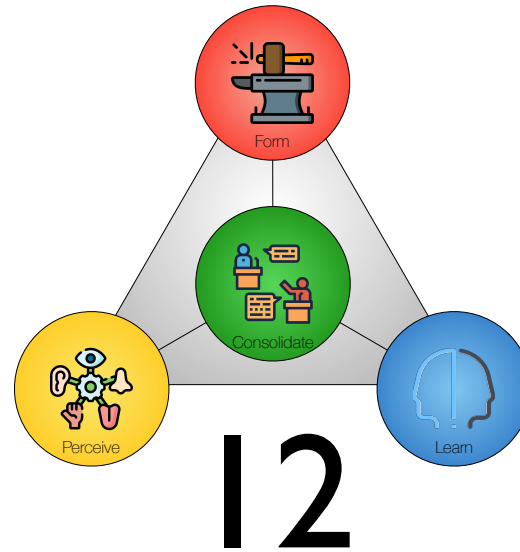


Abstraction layers **conceal** the operational details of external components and allow for **replacing** one external element with another or an internal component.



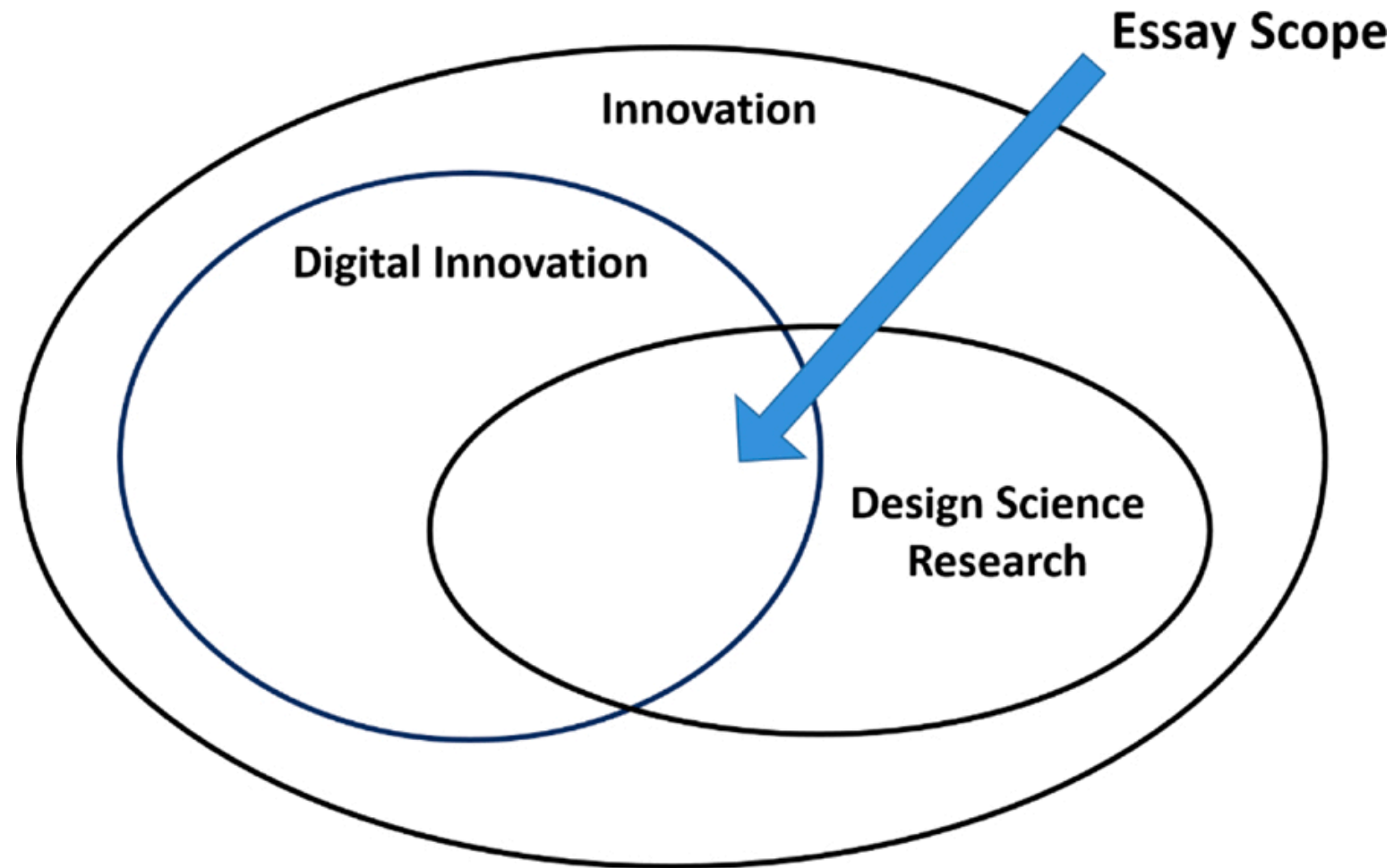
Communicating vessels





Innovation as Design Science

Scope of Essay – Digital Innovation and Design Science Research



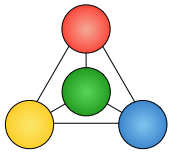
Main Approach: Effectuation

- *Effectuation processes* take a set of *means* as given and focus on selecting between possible *effects* that can be created with that set of means. *Causation processes* take a particular *effect* as given and focus on selecting between *means* to create that effect.
- Effectuation logic has some commonality with the idea of “*need-solution*” *pairs* advanced by von Hippel and von Krogh, who argue that problem formulation need not precede problem solving, but rather *a need* and *a solution* may be *discovered* together.

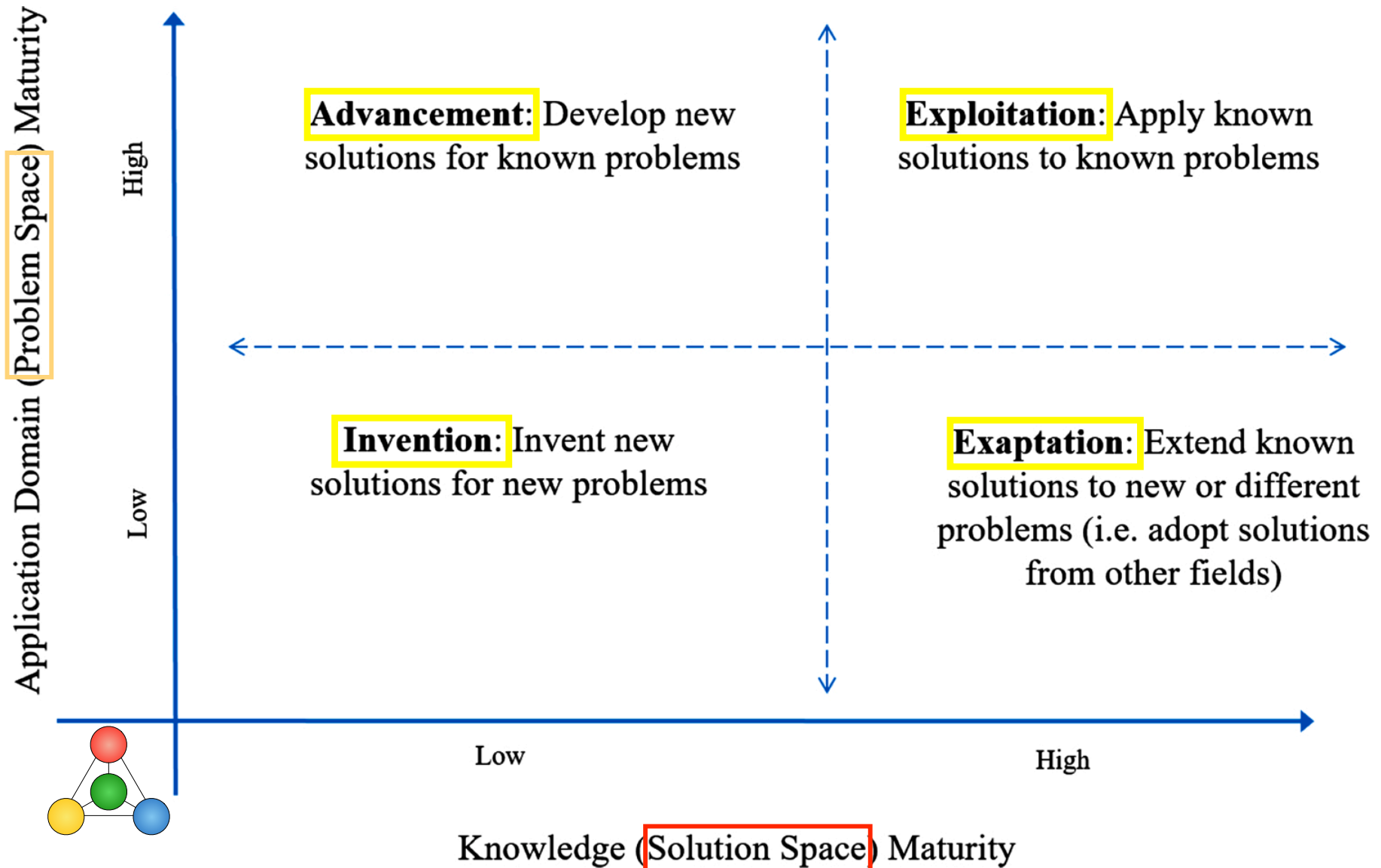
Knowledge Innovation

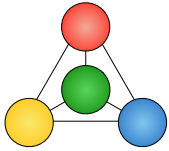
Maturity Dimensions

- The *application domain* (**problem**) maturity dimension, which resonates with the key roles in innovation of opportunities, tasks and problems, markets, needs and fields.
- The *knowledge* (**solution**) maturity dimension, which resonates with the key roles in innovation of new ideas, new insights, new knowledge and skills, technological know-how, new knowledge, and learning.

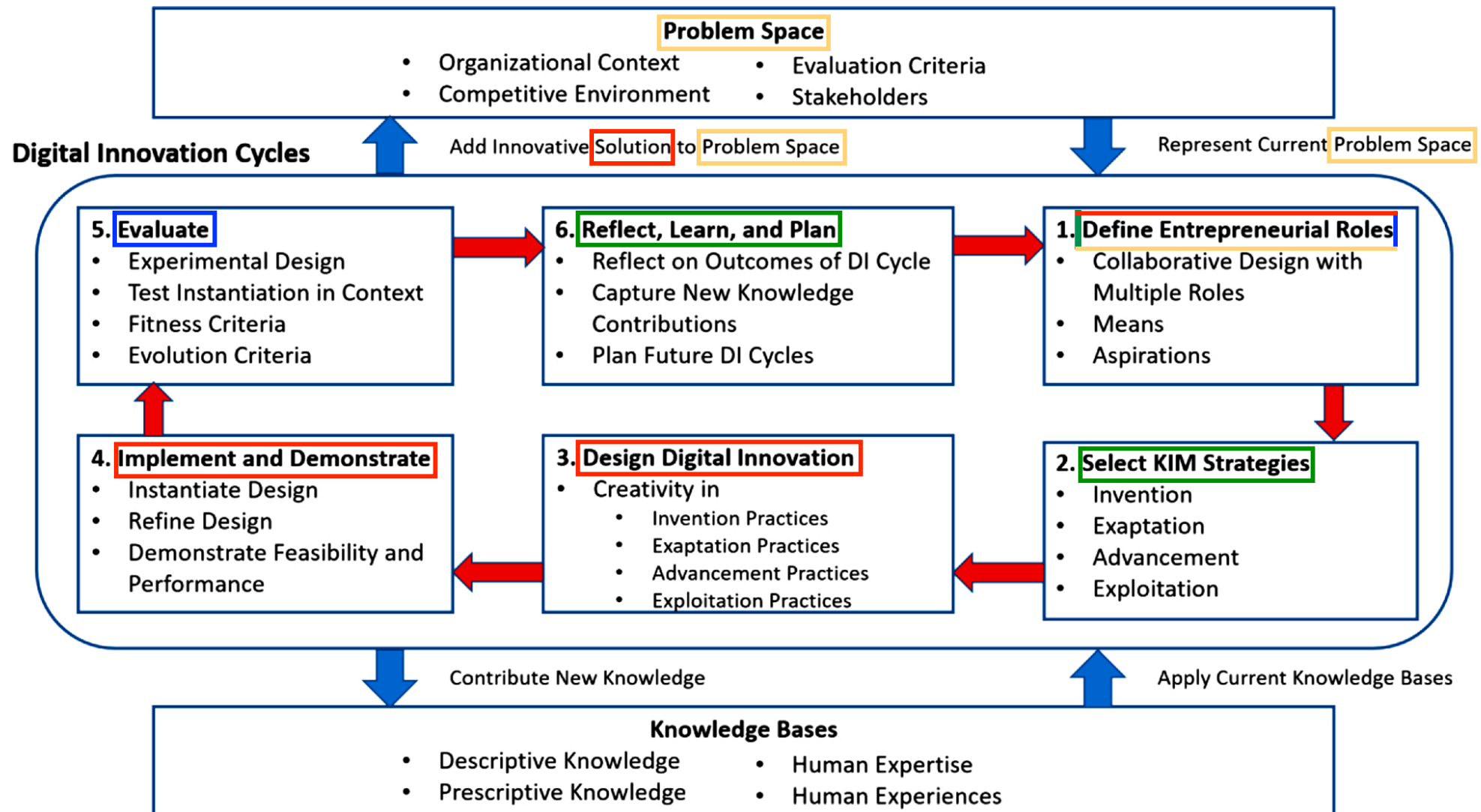


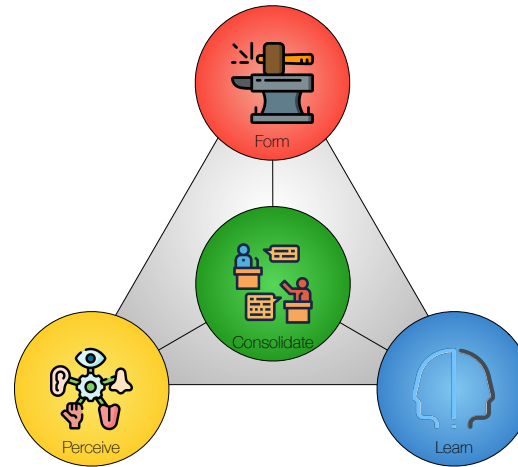
KIM: Knowledge Innovation Matrix





Process model with strategies and roles





Mini-Project and Exam

Mini-project Work Day

May 6

- No lecture.
- You can buzz me in **Teams** or see me in my **office** (5.2.15) between **8.30 – 12**.

Mini-project deadline: May 11

Mini-Project

- The exercises are meant to contribute to your semester project **and** the mini-project.
- Contributions: Consistent reasoning, exploration of alternatives, decisions on scope, leverage points, hypotheses about the solution, ...
- Work alone or in groups.
- **Technically, the mini-project is not a deliverable and will not be graded.**

See Moodle for detailed and updated description

Exam (with mini-project)

- The mini-project is **not graded** but serves as a **basis** for the oral exam.
- You will give a five-minute **presentation** on the mini-project. Following this, the exam will cover **any part** of the syllabus.
- Preferably, the **discussion** will be related to the mini-project.
- You may use **examples** and **illustrations** from the mini-project during the oral exam.

See Moodle for detailed and updated description

Exam (no mini-project)

- You will randomly draw a question **related to the topics from the exercises**.
- You will give a five-minute **presentation** about the selected topic. The exam will then cover **any part** of the syllabus.

See Moodle for detailed and updated description